

CSS Fonts

Font Selection is Important

Choosing the right font has a huge impact on how the readers experience a website. The right font can create a strong identity for your brand. Using a font that is easy to read is important. The font adds value to your text. It is also important to choose the correct color and text size for the font.

Generic Font Families

In CSS there are five generic font families:

1. **Serif** fonts have a small stroke at the edges of each letter. They create a sense of formality and elegance.
2. **Sans-serif** fonts have clean lines (no small strokes attached). They create a modern and minimalistic look.
3. **Monospace** fonts - here all the letters have the same fixed width. They create a mechanical look.
4. **Cursive** fonts imitate human handwriting.
5. **Fantasy** fonts are decorative/playful fonts.

Difference Between Serif and Sans-serif Fonts



Note: On computer screens, sans-serif fonts are considered easier to read than serif fonts.

Some Font Examples

Generic Font Family	Examples of Font Names
Serif	Times New Roman Georgia Garamond
Sans-serif	Arial Verdana Helvetica
Monospace	Courier New Lucida Console Monaco
Cursive	<i>Brush Script MT</i> <i>Lucida Handwriting</i>
Fantasy	COPPERPLATE Papyrus

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The CSS font-family Property

In CSS, we use the `font-family` property to specify the font of a text.

The `font-family` property should hold several font names as a "fallback" system, to ensure maximum compatibility between browsers/operating systems. Start with the font you want, and end with a generic family (to let the browser pick a similar font in the generic family, if no other fonts are available). The font names should be separated with comma.

Note: If the font name is more than one word, it must be in quotation marks, like: "Times New Roman".

Example

Specify some different fonts for three paragraphs:

```
<!DOCTYPE html>
<html>
<head>
<style>
.p1 {
  font-family: "Times New Roman", Times, serif;
}

.p2 {
  font-family: Arial, Helvetica, sans-serif;
}

.p3 {
  font-family: "Lucida Console", "Courier New", monospace;
}
</style>
</head>
<body>

<h1>CSS font-family</h1>
<p class="p1">This is a paragraph, shown in the Times New Roman font.</p>
<p class="p2">This is a paragraph, shown in the Arial font.</p>
<p class="p3">This is a paragraph, shown in the Lucida Console font.</p>

</body>
</html>
```

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CSS Web Safe Fonts

Web safe fonts are fonts that are universally installed across all browsers and devices.

Fallback Fonts

However, there are no 100% completely web safe fonts. There is always a chance that a font is not found or is not installed properly. Therefore, it is very important to always use fallback fonts. This means that you should add a list of similar "backup fonts" in the `font-family` property. If the first font does not work, the browser will try the next one, and the next one, and so on. Always end the list with a generic font family name.

Example

Here, there are three font types: Tahoma, Verdana, and sans-serif. The second and third fonts are backups, in case the first one is not found.

```
p {  
font-family: Tahoma, Verdana, sans-serif;  
}
```

Best Web Safe Fonts for HTML and CSS

The following list are the best web safe fonts for HTML and CSS:

- Arial (sans-serif)
- Verdana (sans-serif)
- Helvetica (sans-serif)
- Tahoma (sans-serif)
- Trebuchet MS (sans-serif)
- Times New Roman (serif)
- Georgia (serif)
- Garamond (serif)
- Courier New (monospace)
- Brush Script MT (cursive)

Note: Before you publish your website, always check how your fonts appear on different browsers and devices, and always use fallback fonts!

Commonly Used Font Fallbacks

Below are some commonly used font fallbacks, organized by the 5 generic font families:

- Serif
- Sans-serif
- Monospace
- Cursive
- Fantasy

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Font Style

The `font-style` property is mostly used to specify italic text.

This property has three values:

- `normal` - The text is shown normally
- `italic` - The text is shown in italics
- `oblique` - The text is "leaning" (oblique is very similar to italic, but less supported)

Example

```
p.normal {  
  font-style: normal;  
}  
  
p.italic {  
  font-style: italic;  
}  
  
p.oblique {  
  font-style: oblique;  
}
```

Font Weight

The `font-weight` property specifies the weight of a font:

Example

```
p.normal {  
  font-weight: normal;  
}  
  
p.thick {  
  font-weight: bold;  
}
```

Font Variant

The `font-variant` property specifies whether or not a text should be displayed in a small-caps font. In a small-caps font, all lowercase letters are converted to uppercase letters. However, the converted uppercase letters appear in a smaller font size than the original uppercase letters in the text.

Example

```
p.normal {  
  font-variant: normal;  
}  
  
p.small {  
  font-variant: small-caps;  
}
```

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Font Size

The `font-size` property sets the size of the text.

Being able to manage the text size is important in web design. However, you should not use font size adjustments to make paragraphs look like headings, or headings look like paragraphs.

Always use the proper HTML tags, like `<h1>` - `<h6>` for headings and `<p>` for paragraphs.

The font-size value can be an absolute, or relative size.

Absolute size:

- Sets the text to a specified size
- Does not allow a user to change the text size in all browsers (bad for accessibility reasons)
- Absolute size is useful when the physical size of the output is known

Relative size:

- Sets the size relative to surrounding elements
- Allows a user to change the text size in browsers

Note: If you do not specify a font size, the default size for normal text, like paragraphs, is 16px (16px=1em).

Set Font Size with Pixels

Setting the text size with pixels gives you full control over the text size:

Example

```
h1 { font-size: 40px; }
```

```
h2 { font-size: 30px; }
```

```
p { font-size: 14px; }
```

Tip: If you use pixels, you can still use the zoom tool to resize the entire page.

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Set Font Size with Em

To allow users to resize the text (in the browser menu), many developers use em instead of pixels.

1em is equal to the current font size. The default text size in browsers is 16px. So, the default size of 1em is 16px.

The size can be calculated from pixels to em using this formula: $pixels/16=em$

Example

```
h1 {  
  font-size: 2.5em; /* 40px/16=2.5em */  
}  
  
h2 {  
  font-size: 1.875em; /* 30px/16=1.875em */  
}  
  
p {  
  font-size: 0.875em; /* 14px/16=0.875em */  
}
```

In the example above, the text size in em is the same as the previous example in pixels. However, with the em size, it is possible to adjust the text size in all browsers.

Unfortunately, there is still a problem with older versions of Internet Explorer. The text becomes larger than it should when made larger, and smaller than it should when made smaller.

Use a Combination of Percent and Em

The solution that works in all browsers, is to set a default font-size in percent for the <body> element:

Example

```
body { font-size: 100%; }  
  
h1 { font-size: 2.5em; }  
  
h2 { font-size: 1.875em; }  
  
p { font-size: 0.875em; }
```

It shows the same text size in all browsers, and allows all browsers to zoom or resize the text

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Responsive Font Size

The text size can be set with a **vw** unit, which means the "viewport width".

That way the text size will follow the size of the browser window:

Example

```
<h1 style="font-size:10vw">Hello World</h1>
```

Viewport is the browser window size. 1vw = 1% of viewport width. If the viewport is 50cm wide, 1vw is 0.5cm.

Google Fonts

If you do not want to use any of the standard fonts in HTML, you can use Google Fonts. Google Fonts are free to use, and have more than 1000 fonts to choose from.

How To Use Google Fonts

Just add a special style sheet link in the `<head>` section and then refer to the font in the CSS.

Example

Here, we want to use a font named "Sofia" from Google Fonts:

```
<head>
<link rel="stylesheet" href="https://fonts.googleapis.com/css?family=Sofia">
<style>
body {
  font-family: "Sofia", sans-serif;
}
</style>
</head>
```

Result:

Sofia Font

Lorem ipsum dolor sit amet.

123456790

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Font Pairing Rules

Here are some basic rules to create great font pairings:

1. Compliment

It is always safe to find font pairings that complement one another.

A great font combination should harmonize, without being too similar or too different.

2. Use Font Super families

A font superfamily is a set of fonts designed to work well together. So, using different fonts within the same superfamily is safe.

For example, the Lucida superfamily contains the following fonts: Lucida Sans, Lucida Serif, Lucida Typewriter Sans, Lucida Typewriter Serif and Lucida Math.

3. Contrast is King

Two fonts that are too similar will often conflict. However, contrasts, done the right way, brings out the best in each font.

Example: Combining serif with sans serif is a well-known combination.

A strong superfamily includes both serif and sans serif variations of the same font (e.g. Lucida and Lucida Sans).

4. Choose Only One Boss

One font should be the boss. This establishes a hierarchy for the fonts on your page. This can be achieved by varying the size, weight and color.

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The CSS Font Property

To shorten the code, it is also possible to specify all the individual font properties in one property.

The `font` property is a shorthand property for:

- `font-style`
- `font-variant`
- `font-weight`
- `font-size/line-height`
- `font-family`

Note: The `font-size` and `font-family` values are required. If one of the other values is missing, their default value are used.

Example

Use `font` to set several font properties in one declaration:

```
p.a {  
  font: 20px Arial, sans-serif;  
}  
  
p.b {  
  font: italic small-caps bold 12px/30px Georgia, serif;  
}
```

All CSS Font Properties

Property	Description
font	Sets all the font properties in one declaration
font-family	Specifies the font family for text
font-size	Specifies the font size of text
font-style	Specifies the font style for text
font-variant	Specifies whether or not a text should be displayed in a small-caps font
font-weight	Specifies the weight of a font